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MCF-7 BREAST CANCER CELLS WITH REDUCED DNA REPAIR GENE EXPRESSION

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Abstract :

Pomegranate extract (PE) inhibits the proliferation of breast cancer cells and stimulates apoptosis in MCF-7 breast most cancers cells. While PE is a mighty antioxidant, the present research has been performed to study the mechanisms of movement of PE beyond antioxidation through studying mobile and molecular mechanisms underlying breast tumorigenesis. PE inhibited mobile boom with the aid of inducing cell cycle arrest in G2/M accompanied with the aid of the induction of apoptosis. In evaluation, antioxidants N-acetylcysteine and Trolox did no longer affect cellular increase at doses containing equal antioxidant ability as PE, suggesting that growth inhibition via PE cannot entirely be attributed to its excessive antioxidant ability. DNA microarray evaluation found out that PE downregulated genes associated with mitosis, chromosome organisation, RNA processing, DNA replication and DNA repair, and upregulated genes involved in law of apoptosis and mobile proliferation. Both microarray and quantitative RT-PCR indicated that PE downregulated crucial genes worried in DNA double strand smash (DSB) repair by homologous recombination (HR), together with MRE11, RAD50, NBS1, RAD51, BRCA1, BRCA2, and BRCC3. Downregulation of HR genes correlated with extended stages of their anticipated microRNAs (miRNAs), miR-183 (expected goal RAD50) and miR-24 (anticipated goal BRCA1), suggesting that PE may additionally adjust miRNAs concerned in DNA restore approaches. Further, PE remedy accelerated the frequency of DSBs. These data propose that PE downregulates HR which sensitizes cells to DSBs, boom inhibition and apoptosis. Due to the fact HR represents a unique goal for most cancers therapy, downregulation of HR through PE may be exploited for sensitization of tumors to anticancer tablets.

Keywords : Breast cancer, Homologous recombination, Pomegranate extract

Introduction :

Pomegranate is fed on as whole fruit and also broadly available commercially as fruit juice, wine, or sauce. The fruit may be divided into three elements: the seeds, the juice, and the peel. Extracts of all parts of the fruit appear to have therapeutic residences. Pomegranate fruit is a wealthy source of many phenolic compounds which include flavonoids (anthocyanins, catechins, and other complex flavonoids) and hydrolyzable tannins called ellagitannins (punicalin, pedunculagin, punicalagin, gallagic, and ellagic acid esters of glucose).

The attention of general polyphenols in pomegranate juice (3.8 mg/mL gallic acid equivalents) is higher than different fruit juices (grape, blueberry, black cherry, cranberry, orange, apple, which incorporate most effective 0.4- 2.6 mg/mL of gallic acid equivalents) [1]. In turn, pomegranate juice has the best antioxidant capability in comparison to different polyphenol-rich drinks, which includes red wine, grape juice, blueberry juice, and green tea. The effective antioxidant properties of pomegranate juice were thought to be responsible for numerous health blessings. However, it has now not been hooked up whether antioxidant results per se can provide

an explanation for the moves of pomegranate juice in inhibiting tumor cell boom in vitro [2].

The effects of pomegranate extract (PE) containing primarily ellagitannins have been evaluated in vitro in terms of antiproliferative, antiinvasive, and proapoptotic properties in several different cancer cell lines. Pomegranate juice as a whole has been located to be extra powerful in comparison to remoted phytochemicals given personally. AL although the principle focus of studies has been on prostate cancer; the anticancer capacity of pomegranate has also been examined in breast cancer albeit most effective in in vitro research. Extraordinary PE arrangements reduced growth of each estrogen-receptor effective in addition to estrogen-receptor poor breast cancer cellular strains mammary organ lifestyle MMTV-Wnt-1 mouse mammary most cancers stem cells and decreased the motility and invasion of MDA-231 and SUM 149 cell lines derived from aggressive tumors. The observed results were related to inhibition of NF-κB expression and inhibition of 17β-hydroxysteroid dehydrogenase (aromatase), the enzyme that catalyzes conversion of estrone to estradiol. Estradiol has been cautioned to growth the chance of breast cancer in postmenopausal women leading to hints that dietary interventions that lessen estrogen synthesis may be beneficial for most cancers prevention [3-5].

In summary, these research advised that pomegranate intake can modulate tumor behavior by reducing boom and invasiveness as well as suppressing hormonal carcinogenesis through discount of estrogen tiers and pastime even as the useful houses of pomegranate are bei ng investigated,the studies of its unfavorable effects have not been unnoticed both. Sub chronic toxicity observe in rats confirmed that oral management of PE at 600 mg/kg/d, the best dose tested, for 90 d is without any tremendous unfavorable outcomes based totally on medical observations, ophthalmic exam, frame weights, frame weight gains, feed consumption, clinical pathology opinions, organ weights, hematological and serum chemistry parameters and gross and histopathology analysis after terminal necropsy [6].

The antioxidant potential of pomegranate ellagitannins has been proposed to account for their pharmacological moves, even though the hyperlink between antioxidant ability and tumor boom inhibition has now not been installed. The genomewide transcriptome analysis of PE results in breast most cancers cells have no longer been suggested to date. The gift study, an integrative technique was carried out the usage of Affymetrixgene arrays to pick out modifications in gene expresion following remedy With PE in MCF-7 breast cancer cells and then validating relative mRNA expression with the aid of quantitative RT-PCR (qRT-PCR) [7]. Similarly, in addition evaluation of the expression of expected microRNAs (miRNAs) for substantially modulated genes became executed. To assess whether or not the antioxidant consequences of PE ought to explain the found boom inhibition of breast most cancers cells, cells were uncovered to usual antioxidants, N-acetylcysteine (NAC) and Trolox.

Materials and methods:

Pomegranate extract:

PE is a standardized extract of pomegranate fruit grown in California (Punica granatum L., notable range; Paramount Farms, misplaced Hills, CA). PE is crafted from fruit skins and includes 95% glycone ellagitannins (mono and oligomeric) standardized to punicalagins (37-40%) and unfastened ellagic acid (3.4%) as decided by using excessive-performance liquid chromatography using formerly described techniques [8].

Cell Line, culture conditions, and experimental remedies:

MCF-7 cells had been grown and maintained in minimal essential Medium Eagle (Sigma-Aldrich, St. Louis, MO) supplemented with 25 mM HEPES (Fisher scientific, honest garden, NJ), 20 mM D-(+)glucose,

100 U/mL penicillin and 100 mg/mL streptomycin and five% fetal bovine serum (Sigma-Aldrich). Cells have been maintained at 37.0°C in a humidified atmosphere of 95.0% air and 5.0% CO2. Stock solution of PE (1 mg/mL) became organized in ddH2O. Inventory solution of Trolox (Sigma-Aldrich) (0.5 mM) was organized by way of dissolving 6.26 mg Trolox in 50 mL Na/K buffer and slightly warming the solution in addition (30°C in water tub). Three hundred micromolar working inventory answer of NAC (Sigma-Aldrich) become prepared through serial dilution from a 30 mM stock in ddH2O [9].

Size of feasible cell variety:

The willpower of cellular viability become first examined by way of MTT (3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyl tetrazolium bromide) assay. However, polyphenols have been shown to intervene with the redox chemistry of MTT assay system main to false nice responses. Constant with these observations, PE itself became determined to generate colorimetric responses

inside the absence of cells, probable due to interference with PE polyphenols. Subsequently, the consequences of PE on MCF-7 cells had been assessed by using two opportunity independent cell viability assays, crystal violet assay and acid phosphatase assay. Crystal violet assay offers quantitative facts approximately the relative density of adhering cells while acid phosphatase assay measures feasible mobile numbers with the aid of quantification of cytosolic acid phosphatase interest [10].

In crystal violet assays, way of life medium became eliminated from each properly on the end of treatment duration and cells have been washed with 500 µL PBS. Cells have been constant with 0.1% glutaraldehyde (Fisher scientific) in PBS for 20 min observed by staining with 0.1% crystal violet for 30 min. The crystal violet stain (Sigma-Aldrich) become solubilized in 1% Triton X-a hundred (Sigma-Aldrich) in ddH2O for 30 min and the absorbance was examining with Victor3 V 1420 Multi-Label Counter at 590 nm. Three impartial biological replicates were analyzed each in triplicate.

Trolox Assay:

Trolox equivalent antioxidant capacity (TEAC) assay measures antioxidant strength based on Trolox (6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid) (Sigma-Aldrich) antioxidant ability and is measured in devices known as Trolox equivalents (TEs). Trolox is a hydrophilic vitamin E analog that lacks the phytyl tail and has greater antioxidant capacity because of its accelerated cellular permeability.

Due to the problems in measuring man or woman antioxidant additives of a complicated mixture, Trolox equivalency is used as a benchmark for the antioxidant potential of this sort of combination. The TEAC assay is primarily based on the suppression of the absorbance of radical cations of 2,2'-azinobis (3-ethylbenzothiazoline 6-sulfonate) (ABTS) (Sigma-Aldrich) with the aid of antioxidants inside the test sample when ABTS incubates with a peroxidase (metmyoglobin) and H2O2. The assay was achieved as formerly defined. Briefly, ABTS radical cations had been organized with the aid of including 80 mg of manganese dioxide to 20 mL of 5 mM aqueous inventory solution of ABTS in 75 mM Na/K buffer, pH 7. Trolox turned into used as an antioxidant fashionable [11].

A wellknown calibration curve changed into built for Trolox at 0, 50, 100, 150, 200, 250 and 300 mM concentrations.

Samples of PE and NAC have been serially dilute in Na/K buffer, brought in triplicates in a 96-nicely plate and mixed with 200 mL of ABTS+ radical cation solution. Absorbance turned into read at 750 nm after 5 min in a Labsystems Multiscan MCC/340 microplate reader (Fisher scientific). TEAC values were calculated from the Trolox preferred curve and expressed as TEs (mM). Based totally on the assay results, doses of PE, Trolox and NAC that have been equal in terms of TEs had been used to treat MCF-7 cells to assess the comparative increase inhibition at 72 h. Three independent organic replicates were analyzed each in triplicate.

Float Cytometry:

For flow cytometry studies, MCF-7 cells had been initially plated at a density of 2 × 104 cells/well in triplicates in a volume of 1 mL/well of lifestyle medium in 24-nicely plates. Cells were allowed to stick for 24 h. The next day, media changed into eliminated and replaced with PE-supplemented or PE-unfastened medium. On 24, 48, and 72 h incubation, cells have been harvested by means of trypsinization, three wells were pooled together and organized for drift cytometric analysis of Apo-BrDU and PI staining as according to the protocol (Phoenix float systems, San Diego, CA).

Following trypsinization, cells had been constant with four% formaldehyde in PBS for 30 min on ice observed with the aid of 70% ethanol permeabilization overnight at 20°C. Samples have been enzymatically classified with bromodeoxyuridine triphosphate (BrdU) in TdT response buffer (2.5 mM cobalt chloride and 24 000 U terminal transferase) for 1 h at 37°C to label the 3' -OH ends of fragmented DNA (Roche carried out technological know-how, Indianapolis, IN) [12].

DNA strand breaks have been counterlabeled with FITC-conjugated anti-BrdU monoclonal antibody in line with the manufacturer's recommendations (Phoenix waft structures). Cells were counterstained with 5 µg/mL propidium iodide (Sigma-Aldrich) in the presence of 0.015 U/mL RNase (Roche carried out technology) in PBS for 30 min at room temperature. Samples had been analyzed on BDTM LSR II drift Cytometer (BD Biosciences, San Jose, CA) within 1 h of labeling and at least 10 000 events have been analyzed for each experimental circumstance. Three unbiased organic replicates for every treatment organization had been analyzed [13,14].

Immunofluorescent evaluation of -H2AX Foci:

Cells were propagated in 4-properly Lab-Tek®II Chamber slidesTM (NUNC A/S, Roskilde, Denmark) and treated with exclusive doses of PE. On the cease of every remedy length, cells had been constant with 4% paraformaldehyde, semi-permeabilized with 0.5% Triton-X 100 and washed with PBS as formerly defined. In short, cells were incubated with 1:400 dilution of mouse monoclonal antibody to γ-H2AX (phosphorylated histone 2AX) (EMD Millipore, Billerica, MA), accompanied by 1:200 dilution of goat antimouse antibody to immunoglobulin categorized with fluorescein isothiocyanate (FITC) (Jackson Immunochemicals, West Grove, PA) in PBS containing 10% FBS for 1.5 and 1 h, respectively, and hooked up with Vectashield with 4',6' diamino-2-phenylindol (Vector Laboratories, Burlingame, CA). Cells have been visualized at one hundred the usage of Nikon Eclipse TS100 microscope and analyzed for wonderful nuclear foci. DAPI (1 μg/ mL) in the mounting media became used as a counterstain: as a minimum 100 cells have been counted and cells with more than 4 awesome foci within the nucleus have been considered effective for γ-H2AX. Three unbiased biological replicates had been analyzed [4].

Results:

Increase Inhibition by means of PE:

The consequences of various doses of PE treatment on MCF-7 cellular boom. Remedy with 0-100 μg/mL of PE extensively inhibited MCF-7 mobile increase in a dose- and timedependent way as measured through acid phosphatase assay (determine 1A) and crystal violet assay.

Both assay measurements confirmed comparable results at all of the doses and time factors. A dose of 20 μg/mL inhibited mobile boom with the aid of approximately 30% and 35% at 72 and 96 h, respectively, whereas 50 μg/mL of PE inhibited cell growth by about 50% and 80% at 72 and 96 h, respectively, compared to untreated controls [2]. Cell numbers remained similar at some stage in the treatment length at an exposure dose of 50 μg/mL, indicating the massive inhibition of cell growth. The given that PE is standardized to ∼40% punicalagin and 3.4% unfastened ellagic acid, 50 μg/mL of PE is equivalent to 18.44 μM punicalagin and ∼5.62 mM free ellagic acid, while the relative proportions of different ellagitannins are unknown (figure 1).

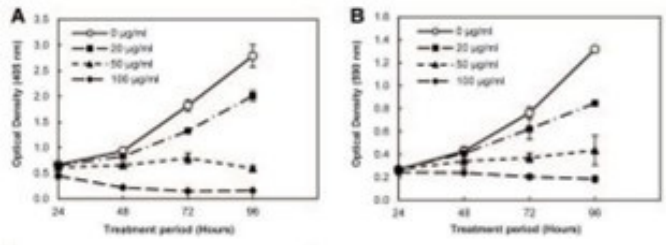

Figure 1: Treatment effects on MCF-7 cell growth.

We also examined the impact of PE pretreatment. Cells maintained in PE-unfastened medium after PE pretreatment showed approximately threefold discount or 48 h put off in cellular boom in assessment to their respective controls. This recommended that PEgene interactions hooked up for the duration of remedy were answerable for the sustained boom arrest

Comparative boom Inhibition of PE, NAC, and Trolox:

We decided whether cellular increase inhibition by way of PE is because of its antioxidant actions. We used TEAC assay to evaluate antioxidant capacity of PE, NAC, and Trolox in terms of TEs. for instance, antioxidant capacity of 50 μg/mL of PE become determined to be equivalent to 40 μg/mL (220 μM) of NAC and 75 μg/mL (280 μM) of Trolox. Boom inhibition of equivalent doses of PE, NAC, and Trolox is shown in parent 1D. In settlement with the outcomes in figure 1A, PE inhibited cell growth in a dose-dependent manner. But, NAC or Trolox at doses containing equal antioxidant capability as PE did not inhibit cell growth [9,12].

This indicated that the growth-inhibitory motion of PE cannot totally be attributed to its antioxidant potential. NAC is understood to inhibit cancer mobile growth but at concentrations which might be 3 orders of significance better (tens of mM) than utilized in our study whilst cytostatic doses of Trolox have not been mounted

Cell Cycle evaluation:

Consistent with the cell viability assay consequences go with the flow cytometric evaluation showed non-stop cellular increase inhibition with 50 μg/mL of PE in the course of the 72h remedy duration. Suggests waft cytometry outcomes for zero and 50 μg/mL of PEtreatment

for 48 h. PE treated cells amassed in G2 segment probable because of G2/M cellular cycle arrest and additionally underwent apoptosis [14].

The degree of cell cycle inhibition became comparable at 24 and 72 h (statistics not proven). These effects suggest that interference with cellular cycle regulation pathways may be one of the mechanisms of cell growth inhibition via PE.

Genome-huge outcomes of PE on MCF-7 Transcriptome:

A total of 903 differentially expressed gene entities (1.5-fold cutoff, $P < 0.05$) had been identified in cells handled with 50 µg/mL of PE for 72 h, where 398 genes have been downregulated and 505 genes were upregulated. DAVID practical annotation analysis, achieved to highlight the maximum relevant move-term enrichment, showed 134 enriched phrases related to downregulated genes and 264 enriched phrases associated with upregulated genes. Top over-represented cellular tactics taken care of by means of rating, –log(P-value).

Twenty-four differentially expressed genes (13 downregulated and eleven upregulated) have been decided on for qRT-PCR validation based on genes classified for most cancers-associated enriched cross-phrases (mitosis, DNA replication, DNA repair, chromosomal employer, chromosomal segregation, regulation of apoptosis, regulation of cell proliferation) and those that displayed the best excessive in differential expression (unbiased of the move-term, based totally on lowest P-values) (figure 2).

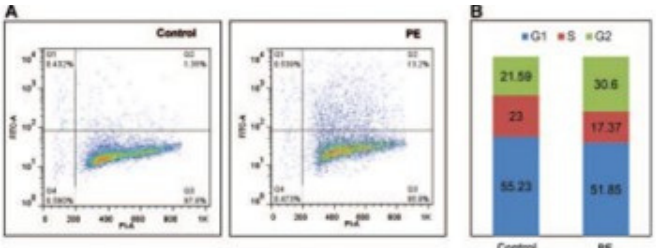

Figure 2: Flow cytometric analysis of PE treatment in MCF-7 cells.

The differential gene expression of most of those genes changed into showed by using qRT-PCR. We targeted on mobile procedures which can explain growth inhibitory and proapoptotic outcomes of PE, which includes DNA harm response, DNA restore and mobile cycle manage DNA harm response and restore genes together with MRE11, RAD50, NBS1, RAD51, BRCA1, BRCA2, BRCC3, and MSH6 that were downregulated in microarrays, showed a reduction in mRNA ranges over a length of 72 h, where the best discount was observed at 72 h. The significance of those genes is underscored by the fact that the proteins encoded by using RAD50, NBS, and MRE11 form the MRE11-RAD50-NBS1 (MRN) complex, which acts as double strand destroy (DSB) sensor, continues genome balance throughout replication, promotes homologous recombination (HR) restore, activates ATM for recruitment to damaged DNA and is essential for mobile viability. RAD51 is a valuable player in maximum HR restore events.

PE remedy resulted in a considerable upregulation of miR-183 and miR-24 however no longer miR-132. Increased miR-183 tiers correlated inversely with mRNA tiers of RAD50 suggesting a possible conversation between miR-183 and RAD50. Similarly, there has been an inverse correlation between miR-24 and mRNA tiers of BRCA1, E2F2, and CCNA2. In precis, this information is suggestive that PE may also upregulate miRNAs involved in DNA repair and cell cycle manipulate however further experimentation is wanted.

Discussion:

The observations within the gift observe display mobile and molecular moves of PE beyond antioxidation in MCF-7 cells which include proof of downregulation of DNA repair.

While in advance research have suggested growth inhibitory, proapoptotic, and antiinvasive homes of PE in exclusive cancer cell lines that is the first study to illustrate that PE can also affect DNA repair pathway required for the survival of cancer cells.

Constant with earlier reviews, our take a look at confirmed that PE remedy inhibited cancer cellular growth in a dose- and time-dependent way, as assessed by two independent mobile increase and viability assays. Growth inhibition changed into associated with the G2/M mobile cycle arrest and the induction of apoptosis. Secondly, PE pretreatment considerably delayed mobile increase. This locating can be of scientific relevance and indicates that intermittent exposure to PE nutritional supplements might be powerful without compromising anticancer sports, which may be deemed as a more possible nutritional approach. Thirdly, comparative increase inhibition assessment of PE, NAC, and Trolox indicates that the growth inhibition by using PE isn't always solely attributed

to its high antioxidant capability.

This turned into a robust reason for investigation of different feasible mechanisms of action, for which we hired Affymetrix microarrays [7].

Microarray analysis confirmed that PE remedy appreciably modulated over 900 genes. The maximum enriched move-phrases among downregulated genes covered mitosis, chromosomal business enterprise, RNA processing, DNA replication, DNA harm response, and DNA repair, while the most enriched GO-terms among upregulated genes protected law of apoptosis and regulation of mobile proliferation. Despite the fact that a lot of those cell procedures might also control cell increase and dying, this observe links for the first time PE publicity and downregulation of DNA restore genes.

Conclusion:

In summary, our have a look at suggests that PE targets a specific DNA restore pathway that is crucial for cancer cellular survival each spontaneously and after drug task. This gives a robust purpose for similarly trying out of PE as a dependable adjunct to the prevention of breast cancer recurrence thru its movements on breast cancer cells. But, in addition research is wanted to demonstrate those effects in animal research and in tissue samples from women with breast cancer.

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Conflict of interest:

Author declares there is no conflict of interest.

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